

Analysis companion: MIDI-to-amplitude diagnostics

This companion defines the fixed-prediction analysis in the manuscript. It supplies cached targets, note annotations, MIDI features, predictions from all eight internal predictors, exact fold membership, analysis code and a file-hash manifest. It is a local submission package; no public repository or public data release is claimed. The supplied NumPy program reproduces the numerical comparisons without training or a GPU. Historical recordings, training checkpoints and a complete historical software environment are not included.

1. Target and data provenance

There are 186 tracks: URMP 136, Bach10 40, PHENICX 2 and TRIOS 8. Piece identity and exact amplitude/pitch duplicates define 44 groups. `tracks.json` gives the original track identifiers, corpus, instrument, group, cached hop time and target hash. `folds.json` indexes this ordered track list, assigning training, validation and evaluation roles separately in each of four folds. Every track is evaluated once; no group crosses roles within a fold. Twenty percent of the remaining groups serve as validation. Split seed: 20260911.

Targets are the retained float32 amplitude arrays, without additional evaluation gain fitting or normalization. The archived preprocessing source specifies 16 kHz audio and `librosa.feature.rms` with a 320-sample window and 160-sample hop. That historical call leaves centering and padding at library defaults; its exact historical librosa version has not been recovered. The cached target arrays, rather than a claimed bit-exact reconstruction from recordings, define this study. They are not calibrated sound-pressure levels. Recording gain, room response, microphone placement and source separation can affect this target. A constant positive track gain does not change within-track Pearson correlation, but it does affect linear errors and may affect learning.

The archived corpus loaders select instrument-part files: URMP `AuSep_*`, Bach10 instrument-suffixed WAVs, and instrument-named PHENICX/TRIOS files. They average channels to mono and resample to 16 kHz before RMS extraction; they do not select ensemble mixes. This is source-path evidence, not a new acoustic isolation measurement. Bach10 has onset annotations; its loader constructs each offset from the next onset, with the last extending to the retained horizon. All 1,917 adjacent Bach10 pairs in the cache match that rule. Other loaders retain supplied note durations. These conventions affect interval means and preclude attributing the URMP/non-URMP contrast solely to corpus acoustics. Notes contain absolute onset, offset, pitch and velocity. Velocity is zeroed before extracting every input feature, including features that might otherwise depend on it. No audio-derived velocity, teacher amplitude or audio feature is supplied at inference. Performed timing is supplied, so this is not score-to-performance generation. Note-interval RMS averages describe the observed track envelope; they do not identify isolated-note acoustic energy. Overlap exclusion below checks annotation interference but cannot eliminate reverberation or recording effects.

2. Input features and predictors

The 20 feature coordinates, in zero-based order, are: activity; pitch/127; normalized note phase; $\log(1 + \text{elapsed note time})$; onset proximity; offset proximity; local note density; preceding pitch interval; its absolute size; preceding rest; local pitch trend; rhythmic regularity; three instrument-family indicators; pitch position within track range; phrase position; relative duration; running pitch direction; articulation ratio. Context is computed from the supplied MIDI. Exact windows, clipping, overlap handling and boundary rules are in `source/local_run_20260912/source_snapshot/src/model/solo/dataset.py`, function `notes_to_frame_features`. The supplied feature arrays preserve the evaluated inputs.

Mamba uses model dimension 128, six layers, state size 16, convolution width 4, expansion 1, bidirectionality and dropout .2. S4D uses dimension 256, four layers, state size 64, bidirectionality and dropout .2. Both attach the same decoder design: a frame BiGRU, two Transformer layers with eight heads and feed-forward width 1024, and the existing note-aware amplitude head. Their total parameter counts are 7.66M and 8.59M. The independent standard BiGRU has two bidirectional layers, 256 units per direction and dropout .3; linear projections produce pitch logits and softplus amplitude, with 1.65M parameters. It was added after viewing the original results. This comparison is not an isolated ablation of the note-aware head.

Each neural run uses seed 42, batches of 32 random 512-frame crops, Adam without weight decay, initial learning rate .001, ten warm-up epochs, cosine decay to 1e-6, at most 300 epochs and patience 60. Selection

minimizes validation loss on eight deterministic, equally spaced crops per track. Evaluation runs full tracks. Exact commands/configuration and model source are included under `source`; they document the frozen runs rather than a newly executed replication.

The loss is voiced-frame pitch cross-entropy plus twice the amplitude loss. With natural-log amplitude and epsilon $1e-7$, amplitude loss combines frame Huber loss (delta 1), voiced-frame correlation loss (weight .3), consecutive-voiced-frame first-difference MSE (.2), and stride-eight mean-pooled log-amplitude MSE (.3). The legacy `per_crop_amp_norm` flag subtracts the SAME target voiced-log mean from prediction and target; it does not fit separate prediction gain. Exact masking and empty-crop rules are in `compute_baseline_loss`. All other optional loss terms are disabled in the frozen recipes.

Five references use the same folds. The gate is one on the union of annotated note frames and zero elsewhere; it has no fitted level. Global OLS and instrument OLS fit an intercept and the 20 frame features to $\log(\max(\text{target}, 1e-7))$ on training frames. Instrument OLS fits one regression per instrument, with global fallback. Note OLS regresses $\log(\max(\text{note mean}, 1e-7))$ on mean note features using training notes of at least four frames. Exponentiated predictions use the fixed numerical log bounds $[-30, 10]$. No bound was active in the retained outputs. Shared envelopes multiply those note levels by instrument-specific templates averaging mean-normalized training notes in 64 phase bins. Templates are interpolated and renormalized at the requested duration. Rendered notes of at least one frame contribute; overlapping contributions are averaged and unannotated frames are zero. These references test timing, feature-based levels and reusable shapes, not the complete score-feature basis model of prior performance research.

3. Metrics, coverage and uncertainty

All reported means weight tracks equally. Pearson correlation is undefined with fewer than three samples or standard deviation below $1e-10$ in either array. Undefined values remain missing and their denominators are retained. Full-track correlation uses the complete cached horizon; active-frame correlation uses the union of annotated intervals.

For a note with times s, e and cached hop h , its frame interval is $[\max(0, \text{int}(s/h)), \min(T, \text{int}(e/h))]$. The original integer truncation and cached floating-point h are preserved. Eligible primary notes have at least four frames. Across-note correlation compares the predicted and target means of these intervals. Within-note shape first correlates frames within each eligible note, then averages defined note correlations within the track. These two measures are not additive components of the full-track coefficient. Note-wise affine invariance refers to each interval's correlation, not to a physically separable decomposition of an overlapping waveform.

For domain/scale evaluation, apply either identity or $\ln(a+1e-7)$, then Gaussian smoothing to both prediction and target. Standard deviations are 0, .05, .1, .2, .5 and 1 seconds, converted using the cached hop; boundary mode is reflect and truncation is four standard deviations. Zero smoothing is identity. No prediction is refitted for a measurement condition. The main result uses all scales; no best-scale model selection is performed.

In the frozen legacy report, `rmse_log` means full-track log RMSE and `rmse_log_active` means active-frame log RMSE. In the new display/export dictionary these are explicitly named `full_log_rmse` and `active_log_rmse`; original report fields are retained unchanged for provenance. Linear RMSE is $\sqrt{\text{mean}((\text{prediction}-\text{target})^2)}$; log RMSE replaces both arrays by $\ln(a+1e-7)$. Compute both on full tracks and active frames, then average tracks. No test-fitted gain or offset is used. Errors measure agreement in cached recording units, not perceptual quality. The main table uses log RMSE of note means; the four full/active frame-error views are supplied separately. Note OLS and shared envelopes have particularly large full-track log errors because their unannotated-frame predictions are zero while recorded tails need not be. The unit gate is an uncalibrated timing control, explaining its large error.

Paired intervals resample piece/duplicate groups with replacement, keeping every track of each sampled group, and recompute the equal-track mean difference. There are 5,000 draws, seed 20260911, with 2.5th/97.5th percentiles. Pair-specific defined coverage determines the group set. Intervals are pointwise and conditional on these fixed predictions, seed and folds; they do not measure training-seed variation, correct multiple comparisons or remove historical test-based development. They are descriptive precision checks for the studied predictors, not confirmatory tests of newly selected algorithms. A cross-zero interval neither establishes equivalence nor proves absence of improvement.

4. Post-review annotation and population checks

The dated `PROTOCOL.md` records these additional analyses before their computation, after reviewing earlier results. This is not preregistration. Predictions, folds and targets are unchanged. All positive-length annotations contribute to overlap detection, including notes shorter than the primary minimum. For the frame check, exclude an entire eligible note if any of its frames has multiplicity greater than one. For the independent continuous-time check, exclude both whole notes in every overlapping half-open time-interval pair, without quantization. No fragments are trimmed and no retained subset is chosen by scores.

Additional populations are whole tracks with no frame overlap; the previously frozen native single-lane population at the synthesis grid; exact-target deduplication; URMP; and all remaining corpora together. These definitions are distinct. All model means, coverage and planned paired comparisons are in `expected_results.json`, including unfavorable and positive exceptions. The small non-URMP subgroup is heterogeneous, so its contrast does not identify a causal corpus property.

5. Public synthesis systems

The public-system populations and frozen protocol JSON files are included separately. Neither public predictor uses MIDI velocity. The MIDI-DDSP expression input contains note pitch, note duration and instrument ID; original MIDI2Params receives frame pitches and onset/offset flags. Loudness/amplitude conditioning is generated by the respective predictor, not copied from the target recording. The supplied frozen protocol fields and inference adapters document these inputs. Original MIDI2Params uses the author-designated `best_model_cpu_120.pt` predictor and CustomViolin synthesis checkpoint on 46 violin tracks; it is not the MIDI-DDSP authors' reimplementation. Public MIDI-DDSP uses synthesis step 50000 and expression step 5000 on all 186 tracks. Both preserve absolute onsets on the 250 Hz synthesis grid, allocate overlapping notes to voices, and sum complete waveforms at unit gain. There is no score-based track trimming or gain selection. RMS is extracted from final 16 kHz audio, with window 320, hop 160, centered constant padding and the frozen target horizon. Seeds, tails, reverb and checkpoint hashes are in the supplied protocol files.

Public URMP pretraining may overlap the evaluation pieces. Neither pipeline was retrained on the internal folds. Public-system results describe fixed adapted pipelines and are not evidence of a fair architecture ranking or clean generalization. The companion also contains the already-collected final RMS arrays for all 186 MIDI-DDSP and 46 original MIDI2Params tracks, mapped to the same cached targets. Each source RMS file is checked against its original completion-record SHA-256. `verify_public_rms.py` uses NumPy/SciPy to recompute full, active, note-mean and within-note correlations plus log correlation and one-second log-smoothed correlation. It checks 1,392 per-track values, 36 public-subset means and 120 primary paired-bootstrap intervals against the frozen reports. These checks reproduce evaluation from archived final RMS, not waveform generation or RMS extraction from audio. No new prediction, gain fit, track selection or inference is performed.

6. Reproduction

Use Python with NumPy (verified with NumPy 2.4.3). From the extracted companion directory run `python verify.py`. It independently recomputes correlations, errors, overlap masks, 576 planned paired-bootstrap comparisons and fold group separation from the supplied arrays, checking against the stored results at tolerance $1e-8$. It uses no GPU and does not read remote files. The independent implementation uses difference-array overlap counting and sorted-interval overlap detection rather than the original audit loops. `verification.json` reports its actual result. The additional `verify_level_audit.py` independently checks 4,464 note-error entries and 45 paired error/aggregation intervals using cumulative-sum interval means and explicit resampled group-row concatenation; its maximum difference is $1.78e-15$. It does not import the original analysis implementation. `manifest.json` identifies the exact packaged files by SHA-256. Source files are supplied to clarify the original computation, including legacy options not enabled in this study; the frozen recipe files determine which options were used.

7. Low-variance interval check

`variance_audit.json` records target standard-deviation quantiles, per-track eligible/defined note-count distributions, and results for all eight predictors at thresholds $1e-8$, $1e-6$ and $1e-4$. No eligible target interval has standard deviation below $1e-6$; just one of 30,352 falls below $1e-4$. Every neural model and the shared envelope has defined shape correlations on all primary eligible notes. Track note counts range from 24 to 777 (median 140.5, quartiles 57.5 and 219). Note OLS has 569 defined correlations across 53 tracks. The stricter $1e-4$ target threshold changes neural shape means by less than $1.4e-5$. Thresholds are in cached recording units, not perceptual thresholds. Arithmetic means of r describe a typical track; Fisher averaging or equal-note weighting would define a different estimand. Neither is silently substituted.

8. Note-level error and alternative aggregation

For each original eligible interval, average prediction and target separately. Note linear RMSE and MAE compare these means; note log RMSE compares $\ln(\text{mean}+1e-7)$, then takes a root mean square across notes. Finally average tracks equally. No test calibration is fitted. The main-table Error field is `note_log_rmse`, distinct from both `full_log_rmse` and `active_log_rmse`; their complete tables are retained. Correlation alone tests co-variation. Linear errors test numerical agreement in recording units; log differences are dimensionless predicted/target note-mean log ratios, with epsilon added before the ratio. They retain overall recording-gain mismatch because no gain is fitted.

The primary arithmetic mean of track correlations describes a typical instrument track without duration/note-count weighting. For sensitivity, average tracks inside each piece/duplicate group then average groups equally; alternatively clip track r to $\pm[1-1e-7]$, take arctanh , average equally over tracks, then apply $\tanh$. In each of 5,000 paired group-bootstrap draws (seed20260911), recompute the full estimator and difference. Fisher aggregation here changes the across-track aggregation, not the preceding arithmetic within-track note average. All planned comparisons and any interval-sign changes appear in the supplementary table and `level_and_aggregation_audit.json`. These exploratory alternatives do not replace the primary estimator. Run `python level_and_aggregation_audit.py` to reproduce them from the same arrays.

For the binary gate at zero smoothing, $\log(\text{gate}+\text{epsilon})$ is a positive affine map of the two gate values. Its Pearson change under the joint log transformation therefore comes from the target transformation. After smoothing, the scale comparison also changes temporal resolution. This is an explanatory property of the specified measurement, not an additional experiment.

Supplementary numerical results

All means weight tracks equally. Intervals are pointwise, conditional on fixed predictions. All planned numerical results, including additional pairs, are in `expected_results.json`.

Absolute error on the complete 186-track population

Predictor	Full linear RMSE	Active linear RMSE	Full log RMSE	Active log RMSE
gate	0.8595	0.9655	5.3607	3.9382
global_ols	0.0275	0.0297	1.0755	0.9690
instrument_ols	0.0264	0.0289	1.0193	0.9151
note_ols	0.0260	0.0280	4.0289	0.9267
note_shape	0.0255	0.0275	4.0196	0.8718
mamba	0.0240	0.0260	0.9136	0.8259
s4d	0.0252	0.0272	0.9357	0.8442
bigru	0.0223	0.0244	0.8692	0.7807

Population and annotation checks

Mamba minus instrument OLS, note-mean correlation. These subsets are sensitivity views of existing tracks, not newly held-out datasets.

Population	Tracks	Groups	Mean difference	95% interval
all186	186	44	0.0302	[-0.0047, 0.0677]
nonoverlap_notes_frame	186	44	0.0131	[-0.0289, 0.0528]
nonoverlap_notes_continuous	186	44	0.0175	[-0.0246, 0.0583]
no_overlap_tracks_frame	133	34	0.0225	[-0.0230, 0.0703]
native_single_lane95	95	27	0.0355	[-0.0080, 0.0808]
deduplicated163	163	44	0.0328	[-0.0007, 0.0685]
URMP	136	27	0.0052	[-0.0347, 0.0464]
non_URMP	50	17	0.0983	[0.0462, 0.1548]

All model views after continuous-time overlap exclusion

28,077 eligible notes remain; exclude both whole notes in every overlapping annotation pair. Full and active views retain the original horizon/mask.

Predictor	Full	Active	Note means	Within notes
gate	0.4420 (n=186)	undefined	undefined	undefined
global_ols	0.4978 (n=186)	0.2037 (n=186)	0.1373 (n=186)	0.2990 (n=186)
instrument_ols	0.5104 (n=186)	0.2841 (n=186)	0.2417 (n=186)	0.3248 (n=186)
note_ols	0.4756 (n=186)	0.1431 (n=186)	0.1738 (n=186)	undefined
note_shape	0.5488 (n=186)	0.3368 (n=186)	0.1738 (n=186)	0.5227 (n=186)
mamba	0.6167 (n=186)	0.4473 (n=186)	0.2591 (n=186)	0.5993 (n=186)
s4d	0.6071 (n=186)	0.4298 (n=186)	0.2412 (n=186)	0.5856 (n=186)
bigru	0.5993 (n=186)	0.4258 (n=186)	0.2200 (n=186)	0.5993 (n=186)

Verification

An independent NumPy implementation checked 44,640 per-track metric entries, 576 paired bootstrap comparisons, overlap counts and fold-role separation. Maximum discrepancy: 3.45e-15 (tolerance 1e-8). Original primary metric reproduction maximum discrepancy: 1.78e-15. Input hashes are retained in the report. These checks establish numerical consistency, not unseen-data validity.

Individual corpus checks

Mamba minus instrument OLS on note means. These are exploratory comparisons; Bach10 constructs offsets from the next onset, so interval conventions differ. Very small group counts provide limited uncertainty information.

Corpus	Tracks	Groups	Difference	95% interval
Bach10	40	10	0.0896	[0.0373, 0.1434]
PHENICX	2	2	0.0566	[-0.0399, 0.1532]
TRIOS	8	5	0.1523	[-0.0437, 0.3692]
URMP	136	27	0.0052	[-0.0347, 0.0464]

Within-note nonconstancy checks

Of 30,352 eligible target intervals, none has standard deviation below $1e-8$ or $1e-6$; one is below $1e-4$. Minimum: $7.2617e-5$; first percentile: .00075994; median: .00589944 (cached RMS units). Per-track eligible counts: min 24, Q1 57.5, median 140.5, Q3 219, max 777. Neural predictors and shared envelopes have defined correlations on all eligible notes. Note OLS: 569 notes in 53 tracks.

Predictor	Original within-note mean	Target SD $\geq 1e-4$	Change
gate	undefined	undefined	undefined
global_ols	0.298925	0.298931	0.0000058
instrument_ols	0.320473	0.320485	0.0000125
note_ols	0.045612	0.045612	0.0000000
note_shape	0.516908	0.516919	0.0000110
mamba	0.592058	0.592071	0.0000125
s4d	0.580713	0.580721	0.0000074
bigru	0.594726	0.594739	0.0000133

All means are identical to the original at the two intermediate thresholds. Full per-track results and planned thresholds are supplied; these checks do not turn correlations into a calibrated perceptual accuracy scale.

Note-mean numerical errors

Compute each note mean first, then its linear or log error, average squared/absolute errors over notes and take the root where appropriate; finally average tracks equally. The log uses epsilon $1e-7$ after averaging, without fitted gain. Main-table Error is note-mean log RMSE. Full/active frame errors above remain reported separately.

Predictor	Note linear RMSE	Note log RMSE	Note linear MAE
gate	0.963498	3.708426	0.963335
global_ols	0.027222	0.814200	0.023670
instrument_ols	0.026170	0.745229	0.022023
note_ols	0.024914	0.723253	0.021389
note_shape	0.024916	0.723343	0.021391
mamba	0.023161	0.688952	0.019876
s4d	0.024761	0.718275	0.021263
bigru	0.021582	0.649831	0.017626

Paired difference	Error measure	Difference	95% interval
mamba minus instrument_ols	note_linear_rmse	-0.003009	[-0.006230, 0.000115]
mamba minus instrument_ols	note_log_rmse	-0.056276	[-0.133192, 0.022958]
mamba minus instrument_ols	note_linear_mae	-0.002147	[-0.005260, 0.000902]
mamba minus note_shape	note_linear_rmse	-0.001755	[-0.004655, 0.000852]
mamba minus note_shape	note_log_rmse	-0.034391	[-0.106403, 0.034737]
mamba minus note_shape	note_linear_mae	-0.001514	[-0.004441, 0.001121]
mamba minus bigru	note_linear_rmse	0.001579	[-0.001146, 0.004986]
mamba minus bigru	note_log_rmse	0.039121	[-0.045899, 0.140723]

Paired difference	Error measure	Difference	95% interval
mamba minus bigru	note_linear_mae	0.002250	[-0.000509, 0.005726]

Alternative aggregation

Primary: arithmetic mean of per-track r . Group arithmetic: average tracks within group, then groups equally. Track Fisher: arctanh of each clipped track correlation, average tracks, then tanh. Reapply the complete estimator inside each group-bootstrap draw. These do not refit models or define a preferred estimator after observing results.

Comparison / view	Arithmetic tracks: gap [CI]	Arithmetic groups: gap [CI]	Fisher tracks: gap [CI]
mamba minus instrument_ols / raw	0.1063 [0.0858, 0.1302]	0.1126 [0.0873, 0.1394]	0.1031 [0.0826, 0.1274]
mamba minus instrument_ols / active_raw	0.1632 [0.1355, 0.1943]	0.1701 [0.1388, 0.2031]	0.1645 [0.1345, 0.1974]
mamba minus instrument_ols / note_mean	0.0302 [-0.0047, 0.0677]	0.0343 [-0.0051, 0.0783]	0.0170 [-0.0203, 0.0583]
mamba minus instrument_ols / within_note	0.2716 [0.2385, 0.3078]	0.2756 [0.2402, 0.3142]	0.2764 [0.2420, 0.3160]
mamba minus note_shape / raw	0.0679 [0.0469, 0.0912]	0.0684 [0.0474, 0.0902]	0.0637 [0.0444, 0.0864]
mamba minus note_shape / active_raw	0.1105 [0.0877, 0.1320]	0.1103 [0.0871, 0.1328]	0.1126 [0.0882, 0.1358]
mamba minus note_shape / note_mean	0.0961 [0.0633, 0.1258]	0.0826 [0.0510, 0.1145]	0.0925 [0.0577, 0.1243]
mamba minus note_shape / within_note	0.0752 [0.0583, 0.0917]	0.0686 [0.0519, 0.0847]	0.0753 [0.0586, 0.0923]
mamba minus bigru / raw	0.0174 [0.0024, 0.0335]	0.0209 [0.0079, 0.0355]	0.0146 [0.0000, 0.0302]
mamba minus bigru / active_raw	0.0215 [0.0025, 0.0402]	0.0267 [0.0088, 0.0448]	0.0213 [0.0009, 0.0416]
mamba minus bigru / note_mean	0.0438 [0.0060, 0.0808]	0.0551 [0.0196, 0.0896]	0.0401 [-0.0010, 0.0806]
mamba minus bigru / within_note	-0.0027 [-0.0195, 0.0104]	-0.0069 [-0.0177, 0.0034]	-0.0030 [-0.0188, 0.0098]

The Mamba/instrument-OLS and Mamba/shared-envelope interval signs are unchanged across all four views. The Mamba/BiGRU note-mean interval becomes cross-zero under Fisher aggregation (-.0010 to .0806), while positive under arithmetic-track/group aggregation. The Fisher full-track lower endpoint is only 0.00000477, not evidence for a stable architecture ranking.